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An example notebook

If you want to see an example for documented code, check out [a notebook](#) I wrote based on what we are doing in this tutorial.

The link above leads you to an example for:

- How could I use github to make my code available to others
- How a “code book” could look like. The example you see in the folder is provided as a Quarto markdown file [here](#) and for convenience the report was also rendered as a [html](#) to make it easier to read for potential collaborators. To view the HTML report, you need to download it first.

The example is based on the data you analyse throughout this tutorial, so the actual code might only make sense after you have finished the tutorial. Feel free to revisit this page after you have written some code yourself.

Please note that this is just an example to get you started and such a report might look different depending on your needs.

Below you find some general hints for what to put into different sections that you can see in the qmd file.

YAML headers

```
---
title: "My report for analysis X (project title)"

author: Nina Dombrowski
date: 12/18/2023

format:
  html:
    embed-resources: true

toc: true
toc-depth: 2

execute:
  eval: false

engine: knitr
---
```

The text between the `---` is what is called a YAML header and is completely optional but useful if you document code in Rmarkdown or Quarto.

It can be used to specify:

- The characteristics of your document, i.e. title, authors, date, etc.
- Arguments that control how you render your document, for example in this example to HTML
- Other parameters for your report. For example, I do not want code cells to be executed (eval: false)

Settings (change these before you run the workflow)

```
# Path to working directory
wdir="$HOME/personal/data_analysis"

# Custom variables
download_link="https://raw.githubusercontent.com/ndombrowski/cli_workshop/main/data/seq_project.tar.gz"
```

In one of the first sections of my notebooks I add everything that a user that wants to use my workflow has to change and I try to standardize the code below, i.e. I try to write it in a way so that it does not need to be changed. This ensures that another user could just run my code as is without having to edit anything other than this first section. Typically things to add are:

- From where to start the analyses, i.e. the location of the working directory or project folder
- Custom paths, for example for databases that need to be downloaded from the web or repositories
- Custom variables: In the example above I store the location of my project folder in a variable called `wdir`. I use `$HOME` in it instead of typing out my home directory, so the path works for anyone with the same folder structure on Crunchomics. I also store the link to the data in a variable called `download_link`, I then use the variable in the code below to download the data. By doing it this way I have one location in the code another person needs to change the code when for example the path to the data

changes. The code below stays the same since I use the `download_link` variable and not the actual path any more. When writing code try to think ahead and minimize the number of instances where things need to be changed if for example the location to your data changes.

I tend to NOT add the information on how I use `scp` to log into an HPC to keep my user name and login information private.

Looking at the data (sanity checks)

```
# We work with 8 files from 4 samples
wc -l fastq_files.txt
wc -l samples.txt

# We have as many R1 as we have R2 files
ls data/seq_project/*/*R1* | wc -l
ls data/seq_project/*/*R2* | wc -l

# The sequence header looks as expected
zcat data/seq_project/*/*gz | head

# R1 and R2 reads have the same number of reads
mkdir -p results/counts

for fastq_file in $(cat fastq_files.txt); do
    echo "$fastq_file: $(( $(zcat data/seq_project/*/$fastq_file | wc -l) /4 ))"
done > results/counts/read_counts.txt
```

When I first start working with new data, I try to add as many sanity checks as possible to ensure that my data looks good. That way I avoid that I don't notice an issue and run into trouble further down the line. I at the same time understand my data more and learn with how many samples and how much data I am working with.

I also add such sanity checks whenever I modify my data. For example, when I merge individual files into a large file I might count the lines for the individual files and the combined file simply to ensure that I used my wildcards correctly.

Remember: The computer is only as smart as the person using it and will blindly run your commands. Because of that the computer can do unexpected things and you need to account for that.

Using markdown to make notes

```
# Submitted batch job 380627
sbatch scripts/seqkit_trimmed.sh

# We expect one table with a header line and one line per trimmed fastq file (9 lines)
ls -l results/seqkit
wc -l results/seqkit/stats_trimmed.txt

# Same summary statistics as for the raw reads
awk -F'\t' 'NR>1 {reads+=$4; bp+=$5; n++}
END {print "files:", n; print "reads:", reads; print "bp:", bp; printf "mean reads per file: %.1f\n", reads/n; pr
```

Comments on the seqkit output post trimming:

- After cleaning we work with
 - 214,906 out of 217,486 reads (98.8% maintained)
 - 53,826,288 out of 54,493,987 bp (98.8% maintained)
- Information from the fastp html:
 - Little adapter sequence signal (< 0.05%)
 - Read quality drops at the beginning and end of the read -> reconsider re-running fastp and use `--cut_front`, `--cut_front_window_size 4` (default setting) and `--cut_front_mean_quality 20` (default setting). Note that `--cut_front` is disabled by default

In the example above I use markdown to not only document my code but also add some comments whenever I might need it. In the example above, I added some notes about what the results from seqkit stats and the fastp HTML reports actually told me.

This kind of documentation can be useful for:

- Justifying decisions further down the line. In this example, I might decide on how to clean the sequencing data. For example, if I would have found a lot of low quality reads or the adapter still being part of the sequence then I would have specifically cleaned my sequences to deal with that
- Future you. If you read the report a month, or a year, later you have the key information in your report and don't have to open any files or tables that are located elsewhere

Documenting external scripts

```
mkdir -p results/seqkit

# Submitted batch job 380625
sbatch scripts/seqkit.sh

# We expect one table with a header line and one line per fastq file (9 lines)
ls -l results/seqkit
wc -l results/seqkit/stats_raw.txt
cat results/seqkit/stats_raw.txt
```

This part might make more sense after you have worked through the part of the tutorial about using an HPC. But what you see here is how I have written down code that simply says that I submitted a script to a HPC but it does not actually say how I ran the SeqKit software. The actual code is “hidden” inside of the seqkit.sh script. This also means that a person reading your workflow does not have the code right away. You can deal with this in two ways.

1. Instead of the sbatch command, you can add the actual line of code that was run on the compute node.
2. When publishing your code with your manuscript, add the whole scripts folder to where you publish the main code, i.e. on [github](#) or [zenodo](#)

I tend to prefer number 2 because I like to record the code in my notebooks exactly as I ran it but you can do it differently as long as all the code you ran is recorded and accessible to others once you publish your data.

Going further

From notes.md to a Quarto document

Your notes.md and my example notebook are written in the same way: Markdown text with code blocks. The difference is that the example notebook is a **Quarto** document (it ends in .qmd). Quarto turns such a file into a web page (HTML), a PDF or a Word file, with a table of contents and colored code. That makes it easier to share your notebook with others, for example with your supervisor or together with a paper.

If you want to try this in RStudio (recent versions of RStudio come with Quarto):

1. Go to **File**, then **New File**, then **Quarto Document**
2. Copy the content of your `notes.md` into the new file, below the YAML header at the top (see “YAML headers” above)
3. Save the file and click **Render**

Code blocks that you wrote as ````bash`, like in your notes, are only shown on the page, not run. The [Quarto tutorial for RStudio](#) explains the next steps.

Further reading

- [A Guide to Reproducible Code in Ecology and Evolution](#) from the British Ecological Society explains how to organize project folders and how to write clear code that others can rerun. The examples are mostly in R, but the advice applies to bash and other languages as well.
- [Markdown Basics](#) on the Quarto website lists everything else you can do with Markdown, for example lists, links, bold text, images and tables.